Instituto Politécnico Nacional
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Diseño y Análisis de Algoritmos

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TAREA No. 3

1. A binary tree is a rooted tree in which each node has at most two children. Show by induction that in any binary tree the number of nodes with two children is exactly one less than the number of leaves.

   Solution

   Proof. (By induction on the number of nodes)
   Base case. A binary tree consisting of a single node. It has 1 leaf and 0 nodes with 2 children.
   Inductive step. Assume for some binary tree with k-nodes this property holds. Let's say it has l leaves and $l-1$ nodes with two children. If we take away one leave to make a binary tree of size $k-1$, we have two different scenarios:

   - That leaf's parent had only one child.
     When we remove the leaf, its parent will have no descendants and so it becomes a new leaf. Overall, the number of leaves is not affected. Since we also didn't alter any node with two children, then the property holds.
   - The leaf's parent had two children.
     In this case, when we remove the leaf, we reduce the number of leaves in 1, but we also reduce the number of nodes with two children in 1, and the property holds.

   Since in any case the property holds, we have shown that the property is still true for a binary tree with k-1 nodes. □

2. We have a connected graph $G = (V, E)$, and a specific vertex $u \in V$. Suppose we compute a depth-first search tree rooted at u, and obtain a tree T that includes all nodes of G. Suppose we then compute a breadth-first search tree rooted at u, and obtain the same tree T. Prove that $G = T$. (In other words, if T is both a depth-first search tree and a breadth-first search tree rooted at u, then G cannot contain any edges that do not belong to T.)

Solution

Proof. (By contradiction)
Suppose $G \neq T$, since both include every node, that implies there is an edge $e = (v, w)$ in G that is not in T. Since T is a depth-first search tree, v and w should have a descendant relation, as proven in class. That means there are two paths that connect v and w. Hence, there is a cylce in G. However, G is a BFS tree and therefore it doesn't contain cycles. We have reached a contradiction, so our initial supposition is false. □

3. Some friends of yours work on wireless networks, and they're currently studying the properties of a network of n mobile devices. As the devices move around (actually, as their human owners move around), they define a graph at any point in time as follows: there is a node representing each of the n devices, and there is an edge between device i and device j if the physical locations of i and j are no more than 500 meters apart. (If so, we say that i and j are "in range" of each other.) They'd like it to be the case that the network of devices is connected at all times, and so they've constrained the motion of the devices to satisfy the following property: at all times, each device i is within 500 meters of at least $\frac{n}{2}$ of the other devices. (We'll assume n is an even number.) What they'd like to know is: Does this property by itself guarantee that the network will remain connected? Here's a concrete way to formulate the question as a claim about graphs.
Claim: Let G be a graph on n nodes, where n is an even number. If every node of G has a degree at least $\frac{n}{2}$, then G is connected.
Decide whether you think the claim is true of false, and give a proof of either the claim or its negation.

Solution

The claim is true.
Proof (By contradiction)
Suppose G is not connected. That means, at best we have two independent connected components of G. There are two cases:

- Each component has the same number of nodes $\frac{n}{2}$.
  This means a node from any component is connected, at most, with $\frac{n}{2} - 1$ other nodes. But this contradicts the fact that each node has degree $\frac{n}{2}$.

- There's a component with less nodes than the other.
  If we look at the smallest component, we notice it must have at most $\frac{n}{2} - 1$ nodes. That implies, each node is connected to at most $\frac{n}{2} - 2$ other nodes. Therefore any node in this component has a degree $< \frac{n}{2}$, which contradicts the properties

of each node in G.

Since every case we have leads to a contradiction, the initial supposition must be false.

4. There's a natural intuition that two nodes that are far apart in a communication network – separated by many hops – have a more tenuous connection than two nodes that are close together. There are a number of algorithmic results that are based to some extent on different ways of making this notion precise. Here's one that involves the susceptibility of paths to the deletion of nodes.
Suppose that an n-node undirected graph $G = (V, E)$ contains two nodes s and t such that the distance between s and t is strictly greater than $\frac{n}{2}$. Show that there must exist some node v, not equal to either s or t, such that deleting v from G destroys all $s - t$ paths. (In other words, the graph obtained from G by deleting v contains no path from s to t.) Give an algorithm with running time $O(m + n)$ to find such a node v.

Solution

Proof (By contradiction)
Suppose there is no such node v that when deleted destroys all $s-t$ paths. That implies that for every node n in that path, different to the endpoints, there's another path that doesn't include n. In other words, starting from s (or t) there are at least 2 nodes for each distance d before reaching t. That distance is at least $d \leq \frac{n}{2} - 1$ (although it can be greater than that). Taking into account the amount of nodes in G, we have 2 cases:

- If n is odd, the distance from s to t is $\frac{n+1}{2} \geq \frac{n}{2}$, then for each distance $d \leq \frac{n+1}{2} - 1$ there are at least 2 nodes. So G has the nodes s, t and $2\left(\frac{n+1}{2} - 1\right) = (n+1) - 2 = n - 1$ other nodes, so G has $n + 1$ nodes in total, a contradiction.

- If n is even, the distance from s to t is $\frac{n}{2} + 1 \geq \frac{n}{2}$. As mentioned before, for each distance $d \leq \frac{n}{2} + 1 - 1 = \frac{n}{2}$ there are at least 2 nodes. That implies G has the nodes s, t and $2\left(\frac{n}{2}\right) = n$ other nodes. In total, G would have $n + 2$ nodes, a contradiction.

Given that in both cases we get a contradiction, that means our initial supposition must be false. □

Since the crucial element in the previous proof is the distance from any endpoint of the path, is only natural to think about BFS to find such a node v. The only part we need to add is a line that checks whether if the current layer (different from the first one), consists of only one node. If that's the case, then said node is v.